Lecture 15: Regularization and High-Dimensional Inference

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Intro

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- Last lecture: the factor zoo became a high-dimensional problem
- Today: a stats detour before Kozak et al. (**?**) and Feng et al. (**?**)
- This will be more like an ML/Stats lecture
- Goal: understand shrinkage, selection, and valid inference after selection
- A great reference: Elements of Statistical Learning

Why a Stats Detour?

- Suppose you observe $\{(y_1, x_1), \dots, (y_n, x_n)\}$
- $y_i \in \mathbb{R}$ is a target variable, and $x_i \in \mathbb{R}^p$ is a vector of predictors

$$y_i = x_i^\top \beta_0 + \varepsilon_i, \qquad p \text{ can be large relative to } n$$

- Modern empirical work often has many plausible predictors or controls
- Once p is large, OLS becomes unstable or infeasible
- Regularization is the discipline: restrict complexity, keep computation feasible
- Factor zoo $= p$ large to n

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- Factor zoo $= p$ large to n
- Today: everything in the iid, cross-sectional world, but the similar ideas apply to time series and panel data

Flight Plan

- Part I: Ridge, Lasso, and Elastic Net as high-dimensional linear tools
- Part II: Double Machine Learning for causal inference in high dimensions
- Next class: Kozak et al. (**?**) and Feng et al. (**?**)
- These are modern takes on factor models/model selection

Part I: Regularized Linear Models

Setup

$$y = X\beta_0 + \varepsilon, \qquad X \in \mathbb{R}^{n\times p}, \qquad \beta_0 \in \mathbb{R}^p$$

- y: target vector
- X: feature matrix
- p may be comparable to n, or larger
- Main objective: predict well when signal is weak and dispersed
- Classic Machine Learning methods were *not* created for inference
- But they can be still useful for that! This will be Part II

Why OLS Gets Fragile

$$\hat{\beta}^{OLS} = (X^\top X)^{-1} X^\top y, \qquad \mathrm{Var}(\hat{\beta}^{OLS} \mid X) = \sigma^2 (X^\top X)^{-1}$$

- If $p > n$, then $\mathrm{rank}(X^\top X) \leq n < p$: no inverse
- If $X^\top X$ has small eigenvalues, variance blows up in those directions
- Recall: $(X^\top X)^{-1} = QD^{-1}Q^\top$ for $X^\top X = QDQ^\top$
- If $X^\top X$ has small eigenvalues, D^{-1} blows up!
- Good in low dimension; fragile in high dimension

Prediction Risk and the Bias-Variance Tradeoff

- Suppose you estimate the model and receive a new random pair (y^{new}, x_{new})
- What's the expected squared prediction error, conditional on X?

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$$\mathbb{E}\left[(y^{new} - x_{new}^{\top}\hat{\beta})^2 \mid X\right] = \sigma^2 + \mathrm{Bias}(x_{new}^{\top}\hat{\beta})^2 + \mathrm{Var}(x_{new}^{\top}\hat{\beta})$$

- OLS is unbiased when the model is right
- But low bias can come with huge variance
- Regularization accepts some bias to cut variance sharply
- This is already hinting at why naive regularization will screw inference up!

Penalized Least Squares

$$\hat{\beta}(\lambda) \in \arg\min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{2} \|y - X\beta\|_2^2 + P_\lambda(\beta) \right\}$$

- $P_\lambda(\beta)$ controls model complexity
- Larger λ means stronger shrinkage
- Different penalties create different geometry and different estimators
- Important: $\hat{\beta}$ depends on λ, and λ is a tuning parameter, not a fixed constant

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- How to choose λ? Many ways, but cross-validation is common practice

Ridge Regression

$$\widehat{\beta}^{ridge}(\lambda) \in \arg\min_{\beta} \left\{ \frac{1}{2}\|y - X\beta\|_2^2 + \frac{\lambda}{2}\|\beta\|_2^2 \right\}$$

- Penalty is quadratic: $\|\beta\|_2^2 = \sum_{j=1}^{p} \beta_j^2$
- Large coefficients are expensive
- Ridge keeps all regressors, but pushes them toward zero

Ridge: Closed Form

Taking the FOC is both necessary and sufficient for optimality:

$$\nabla_\beta \left[\frac{1}{2}\|y - X\beta\|_2^2 + \frac{\lambda}{2}\|\beta\|_2^2\right] = -X^\top(y - X\beta) + \lambda\beta$$

$$(X^\top X + \lambda I_p)\hat{\beta}^{ridge} = X^\top y \implies \hat{\beta}^{ridge} = (X^\top X + \lambda I_p)^{-1}X^\top y$$

- Adding λI_p regularizes the inverse problem
- Even when $X^\top X$ is singular, $X^\top X + \lambda I_p$ is invertible for $\lambda > 0$
- λ bounds the eigenvalues away from zero, stabilizing the solution

Ridge in the Eigenbasis

If $X^\top X = QDQ^\top$ with $D = \mathrm{diag}(d_1, \dots, d_p)$, then

$$\hat{\beta}^{ridge} = Q(D + \lambda I_p)^{-1} Q^\top X^\top y$$

If OLS exists,

$$\hat{\beta}^{ridge} = Q \, \mathrm{diag}\left(\frac{d_j}{d_j + \lambda}\right) Q^\top \hat{\beta}^{OLS}$$

- Each principal direction is multiplied by $d_j/(d_j + \lambda) \in (0, 1)$
- Weak directions, with small d_j, are shrunk the most

Why Ridge Does Not Select

Under orthogonal regressors, $X^\top X = I_p$, ridge satisfies

$$\hat{\beta}_j^{ridge} = \frac{x_j^\top y}{1+\lambda}, \qquad j = 1, \dots, p$$

- If $x_j^\top y \neq 0$, then $\hat{\beta}_j^{ridge} \neq 0$
- Ridge rescales correlations continuously
- No thresholding rule means no exact zeros, generically
- This is a great model if you believe you have many weak signals in X to predict y

Questions?

Best Subset as the Ideal

- Now let's say you have p regressors but you want to pick a small subset of them
- Maybe because you value parsimony, maybe because you know some are redundant...

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- Recall the ℓ_0 norm counts the number of nonzero coefficients
- Formally: $\|\beta\|_0 = \sum_{j=1}^{p} \mathbb{1}(\beta_j \neq 0)$

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- This is the literal "pick a sparse model" problem
- Statistically attractive, computationally nasty
- How many regressions with k regressors can you run if you have p candidates?

Lasso

- Recall the ℓ_1 norm is $\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|$
- ℓ_1 is a convex proxy for ℓ_0
- Can you show that the ℓ_0 ball is non-convex, while the ℓ_1 ball is convex?

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Welcome to class, Mr. LASSO (Least Absolute Shrinkage and Selection Operator):

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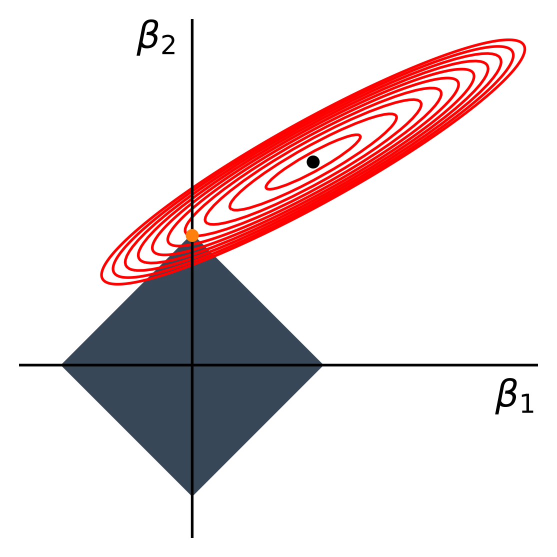
- Still convex, unlike best subset
- Geometry changes, and so does the solution

Lasso Geometry



The criterion can be seen as the Lagrangian of:

$$\hat{\beta}^{lasso}(c) \in \arg\min_{\beta} \left\{ \frac{1}{2} \|y - X\beta\|_2^2 \right\}$$
$$\text{s.t.} \quad \|\beta\|_1 \leq c$$

- For each λ, you can find some c such that the two problems are equivalent
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Soft Thresholding Under Orthonormal Design

If $X^\top X = I_p$ and $z_j = x_j^\top y$, then

$$\hat{\beta}_j^{lasso} = S_\lambda(z_j), \qquad S_\lambda(z) \equiv \operatorname{sign}(z)(|z| - \lambda)_+$$

Equivalently,

$$\hat{\beta}_j^{lasso} = \begin{cases} z_j - \lambda, & z_j > \lambda \\ 0, & |z_j| \le \lambda \\ z_j + \lambda, & z_j < -\lambda \end{cases}$$

- Small correlations get killed exactly
- This is selection, not just shrinkage

Elastic Net

$$\hat{\beta}^{EN} \in \arg\min_{\beta} \left\{ \frac{1}{2}\|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \frac{\lambda_2}{2}\|\beta\|_2^2 \right\}$$

- Lasso part encourages sparsity
- Ridge part stabilizes the problem
- Two penalties, two roles
- Strict generalization, but now you deal with two tuning parameters
- There are fast algorithms for solving these problems – convex analysis rocks!

Why Elastic Net Helps with Correlation

Suppose $x_1 = x_2$ and only $\beta_1 + \beta_2 = c$ matters for fit. Then

$$\beta_1^2 + \beta_2^2 \geq \frac{(\beta_1 + \beta_2)^2}{2} = \frac{c^2}{2}$$

with equality if and only if $\beta_1 = \beta_2 = c/2$.

- For fixed fit, the ℓ_2 term prefers balanced coefficients
- Elastic Net pulls strongly correlated regressors together
- Pure Lasso can keep one and drop the other arbitrarily

Standardization Matters

If $x_j^* = cx_j$, then the same fitted values require $\beta_j^* = \beta_j / c$. For Lasso,

$$\lambda|\beta_j^*| = \lambda \frac{|\beta_j|}{|c|}$$

- The penalty acts on coefficients, not directly on variables
- Rescaling regressors changes the effective penalty if we do nothing
- Standardize predictors before penalizing
- More sophisticated generalizations allow for one λ_j for each β_j

Choosing the Penalty

Standard cross-validation:

- Divide the data into K *random* folds $\mathcal{I}_1, \dots, \mathcal{I}_K$
- Consider a grid $\Lambda = (\lambda_1, \dots, \lambda_m)$
- For each fold k, estimate $\widehat{\beta}^{(-k)}(\lambda)$ on the data excluding fold k

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Cross-validation picks

$$\hat{\lambda} \in \arg\min_{\lambda \in \Lambda} \underbrace{\frac{1}{K} \sum_{k=1}^{K} \underbrace{\sum_{i \in \mathcal{J}_k} \left(y_i - x_i^\top \widehat{\beta}^{(-k)}(\lambda)\right)^2}_{\text{validation error for fold } k}}_{\text{average validation error}}$$

- Choose λ for out-of-sample fit

Questions?

Part II: High-Dimensional Causal Inference

Preliminaries

- Causality still comes from design, not from an algorithm
- Machine Learning won't help if you have a crappy empirical design
- If you have the perfect data, you probably only neeed simple averages
- It's when data is imperfect that you call the firefighters!

Preliminaries

- Causality still comes from design, not from an algorithm
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- If you have the perfect data, you probably only neeed simple averages
- It's when data is imperfect that you call the firefighters!
- Important: ML methods were created for forecasting, not for inference
- This is the Belloni-Chernozhukov move (**?**; **?**; **?**)

Treatment Effects in High Dimensions

$$y_i = d_i\theta_0 + x_i^\top\beta_0 + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid d_i, x_i] = 0$$

- y_i: outcome
- d_i: treatment
- $x_i \in \mathbb{R}^p$: many controls
- Goal: estimate θ_0 and build a confidence interval

Approximate Sparsity

The working assumption is not "truth is literally sparse", but

$$x_i^\top \beta_0 = x_{i,S}^\top \beta_{0,S} + r_i, \qquad |S| = s \ll n, \qquad r_i \text{ small}$$

- A small subset of regressors gives a good approximation
- We *do not* know that subset ex ante
- Regularization is useful only if this approximation is meaningful

Why Naive Lasso Fails

A tempting idea is

$$(\hat{\theta}, \hat{\beta}) \in \arg\min_{\theta,\beta} \left\{ \frac{1}{2}\|y - d\theta - X\beta\|_2^2 + \lambda\|\beta\|_1 \right\}$$

- Even if θ is left unpenalized, missing controls can still bias $\hat{\theta}$
- Selection mistakes matter when omitted controls are correlated with treatment
- Prediction success is not enough for valid inference on θ_0

The Treatment Equation Matters Too

$$d_i = x_i^\top \gamma_0 + u_i, \qquad \mathbb{E}[u_i \mid x_i] = 0$$

- Some controls matter because they predict outcomes
- Some matter because they predict treatment
- Think about the propensity score!
- Omitted variable bias cares about both channels

Post-Double-Selection

Define the selected sets

$$\hat{S}_y = \{j : \hat{\beta}_j^{(y)} \neq 0\}, \qquad \hat{S}_d = \{j : \hat{\gamma}_j^{(d)} \neq 0\}, \qquad \hat{S} = \hat{S}_y \cup \hat{S}_d$$

Procedure:

1. Run Lasso of y_i on x_i
2. Run Lasso of d_i on x_i
3. Run OLS of y_i on d_i and $x_{i,\hat{S}}$

Why the Union Works

- A control creates omitted variable bias only if it matters for the treatment residual, the outcome residual, or both
- The union $\widehat{S}_y \cup \widehat{S}_d$ protects both margins
- If a variable is weak in one equation but strong in the other, the union still keeps it

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$$\hat{S} = \hat{S}_y \cup \hat{S}_d \qquad \text{instead of} \qquad \hat{S}_y \text{ alone}$$

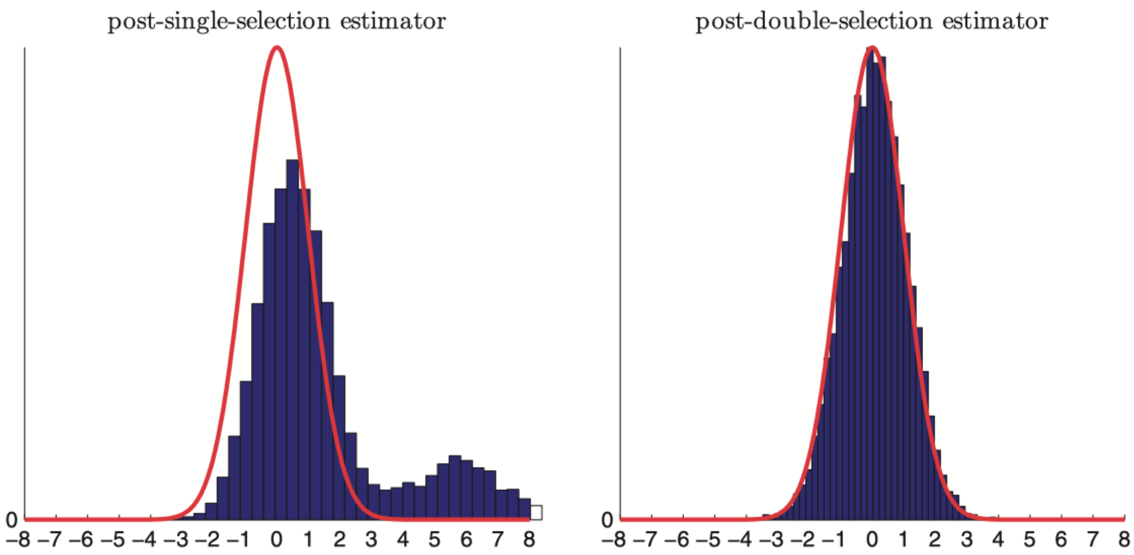
The Main Result

Under approximate sparsity and regularity conditions,

$$\sqrt{n}\left(\hat{\theta}_{PDS} - \theta_0\right) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

- Inference is uniformly valid over a large class of data-generating processes (**?**)
- Confidence intervals are built in the usual way once σ^2 is estimated
- This is much stronger than “it predicts well in simulations”

A Monte Carlo Reassurance



DML Preview

The generalized partially linear model is

$$y_i = d_i\theta_0 + g_0(x_i) + \varepsilon_i, \qquad d_i = m_0(x_i) + u_i$$

A Neyman-orthogonal score is

$$\psi(W_i;\theta,\eta) = (d_i - m(x_i))\,(y_i - g(x_i) - \theta(d_i - m(x_i)))$$

- Estimate g_0 and m_0 with ML on auxiliary folds
- Plug them into the score on held-out data
- Cross-fitting plus orthogonality is the bridge from Double Lasso to DML (**?**)

Questions?

Taking Stock

- Ridge stabilizes
- Lasso stabilizes and selects
- Elastic Net stabilizes and groups correlated signals
- Post-selection inference is not automatic
- Belloni-Chernozhukov show how to recover valid treatment-effect inference with many controls

Readings and References i

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Readings and References ii

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